Current Electricity.

Problem set 1. (on terminal voltage across a battery)

1. A battery of emf '10V' & internal resistance '3Ω' is connected to a resistor. Current in circuit is 0.5A. What is resistance of the resistor? What is terminal voltage of the battery when the circuit is closed?
[17Ω, 8.5V]

2. A voltmeter of resistance 998Ω is connected across a cell of emf 2V and internal resistance 2Ω to measure emf of the cell. Find reading ie P.d across the voltmeter. What is % error in reading of voltmeter?
[1.996V, 0.2%]

3. The potential difference of a cell in an open circuit is 6V which falls to 4V when a current of 2A is drawn from the cell. Calculate e.m.f & internal resistance of the cell.
(6V, 1Ω)

4. The potential difference across a cell is 1.8V when a current of 0.5A is drawn from it. The p.d falls to 1.6V when current 1A is drawn. Find e.m.f & internal resistance of the cell [2V, 0.4Ω].

5. A battery of emf 2V and internal resistance 0.1Ω is being charged by a current of 5A. What will be the direction of current inside battery? What is potential difference between the terminals of the battery. (+ve to -ve terminal, 2.5V)

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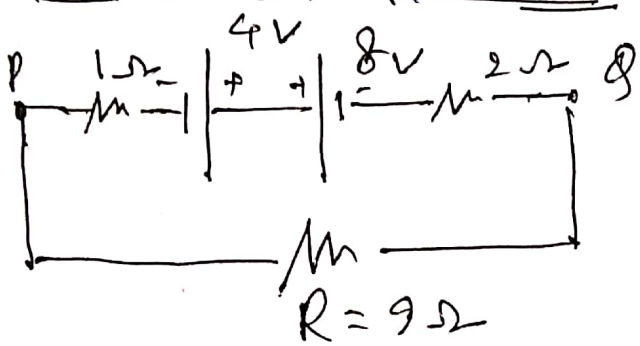
15. 

find current & P.D betn pts P & Q

$\left(\frac{1}{3}A, 3V\right)$

16. For a cell the terminal p.d is 2.2 V when circuit is open and reduces to 1.8 V when cell is connected to a resistance R = 5Ω. find internal resistance (r) of cell? $\left(\frac{10}{9}\Omega\right)$

17. The internal resistance of a 2.1 volt cell which gives a current of 0.2 A through a resistance of 10Ω is

a) 0.2Ω b) 0.5Ω c) 0.8Ω d) 1Ω.

18. A current of 2A flows through a 2Ω resistor when connected across a battery. The same battery supplies a current of 0.5 A when connected across a 9Ω resistor. find internal resistance of the battery. $\left(\frac{1}{3}\Omega\right)$

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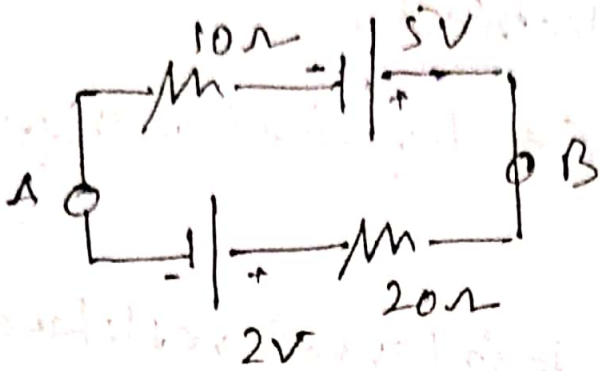
6. 

Find p.d betn. pts A & B. Internal resistance of the cells are negligible.

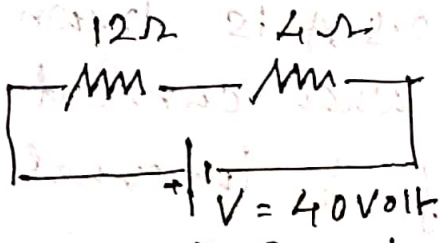
[$V_A - V_B = -4V$
or $V_B - V_A = 4V$]

7. The internal resistance of a 100V battery is 5Ω. Its emf as measured by a voltmeter connected across it shows 20% error. What is resistance of the voltmeter? [20Ω]

Problems on Series Circuit :

8. A fan motor is designed to operate at a current of 2A and at a p.d of 110 V. Find the value of the resistance which should be placed in series with the motor, if the fan is required to operate at 200 V [45Ω]

9. An electric heater of 140Ω resistance and current carrying capacity 1.2A is to be connected with 210 V D.C mains. Find minimum value of additional resistance to be introduced into the circuit. [35Ω in series]

10. 

Find p.d across 12Ω and 4Ω. [30V, 10V]

[ANSWER: P = 88/3]

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(11). Two wires (A) & (B) of diameter 1mm and 2mm are made from two blocks of Cu of mass 1gm each. Find ratio of resistance of wires A & B $\left(\frac{R_A}{R_B} = \frac{16}{1}\right)$

(12) Resistance of a wire of uniform diameter 'd' and length 'l' is 'R'. Find resistance of another wire of the same material but diameter 2d and length 4l [R]

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13 A wire of a certain material is stretched by 10%. Its new resistance and resistivity become respectively

a) 1.2 times, 1.1 times b) 1.21 times, same

c) Both remain the same d) 1.1 times, 1.1 times.

14. The masses of the three wires of Cu are in the ratio of 1:3:5 and their lengths are in the ratio of 5:3:1, find ratio of their electrical resistance.

(125:15:1)

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